

CanvasNet Formal Review Report - Purpose

2026-10-02

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1 Review that README and User Guide are Coherent and Complete

1.1 1. Introduction

1.1.1 1.1 Purpose

This document records the formal review of a set of project files.

1.1.2 1.2 Scope

This is the `Purpose` review-set for the `CanvasNet` repository (the repository's single, hierarchy-wide "north star" review per `reviewmark-usage.md`). It proves that the user-facing documentation — `README.md` and the User Guide (`docs/user_guide/introduction.md`) — is internally coherent and complete, and that it accurately and completely describes the library's actual, current public capabilities, including the PDF-rendering system (`DemaConsulting.CanvasNet.Pdf`) and embedded-font subsystem added on the `feature/pdf-document-support` branch. This is the first formal review ever conducted for this review-set; there is no prior review evidence to supersede.

1.1.3 1.3 Outcomes

Each check must be recorded with one of the following outcomes:

Outcome	Meaning
Pass	The check was performed and the criterion is satisfied
Fail	The check was performed and the criterion is not satisfied
N/A	The check does not apply; justification is required

1.1.4 1.4 Review Details

Field	Value
Project	CanvasNet
Review ID	Purpose
Review Title	Review that README and User Guide are Coherent and Complete
Fingerprint	b76be037bfea1c603ebd0e130abbf5d947f16bb88c7cf30bb27df0f063b6cd2a
Review Date	2026-10-02

1.1.5 1.5 Reviewers

Name	Role	Organization	Signature	Date
Formal Review Agent (Copilot CLI)	Automated Formal Reviewer	DEMA Consulting	(automated — no signature)	2026-10-02

1.1.6 1.6 Files Under Review

File
README.md
docs/user_guide/introduction.md

1.2 2. Review Checklist

1.2.1 2.1 Requirements Checks

Applicable: No

Reason: The review-set contains no requirements files (README.md and the user guide are user-facing technical documentation, not requirements artifacts).

1.2.2 2.2 Design Documentation Checks

Applicable: No

Reason: The review-set contains no design documentation files. `docs/design/introduction.md` is explicitly excluded from this review-set's context in `.reviewmark.yaml` (`"!docs/design/introduction.md"`), by design: the `Purpose` review judges the user-facing promise on its own terms, not against the design documentation that is meant to flow *from* it.

1.2.3 2.3 Technical Documentation Checks

Applicable: Yes

#	Check	Outcome	Justification
DOC-01	Documentation is free of technical inaccuracies	Fail	See Finding F1: the User Guide's <code>PdfDocument.Render</code> exceptions list states that font subtypes <code>/Type0</code> , <code>/Type1</code> , and <code>/Type3</code> are unsupported, contradicting both the current implementation (all three are supported; only <code>/MMType1</code> fails closed) and the User Guide's own preceding prose for the same class.
DOC-02	Documentation is consistent with the current implementation and requirements	Fail	See Findings F1-F3: the User Guide's font-subtype exceptions text (F1) and the missing <code>SystemFontCatalog / TrueTypeFont</code> name-and-style API documentation (F2, F3) leave the documentation inconsistent with (or silent about) implemented, public functionality.
DOC-03	All referenced external documents and dependencies are correctly identified	Pass	README's bundled-font license section references <code>src/DemaConsulting.CanvasNet/Fonts/BundledFonts/OFL.txt</code> , <code>.../BundledFonts/README.md</code> , and <code>.../BundledFonts/NotoFonts-OFL.txt</code> ; all three files exist at the stated paths and the attribution (Red Hat Liberation fonts / Noto Project, both SIL OFL 1.1) is correctly stated.
DOC-04	Documentation is free of spelling and grammar errors	Pass	<code>npx cspell README.md docs/user_guide/introduction.md</code> reports 0 issues (consistent with the repository's cspell dictionary, which already carries the PDF/font-related terms used in both files).

1.2.4 2.4 Code Checks

Applicable: No

Reason: The review-set contains no source code files.

1.2.5 2.5 Logic Error Checks

Applicable: No

Reason: The review-set contains no source code files.

1.2.6 2.6 Error Handling & Logging Checks

Applicable: No

Reason: The review-set contains no source code files.

1.2.7 2.7 Usability / Accessibility Checks

Applicable: Yes

Rationale: Both files exist specifically to make the library's public API usable; their completeness and clarity for that purpose is directly in scope.

#	Check	Outcome	Justification
USE-01	The feature or API is easy to use correctly	Pass	Both files provide clear, minimal, compilable-looking code samples for every major feature (surfaces, codecs, SVG, PDF, fonts, paths, gradients), including a dedicated "Handling Unsupported Features" example showing the <code>CanDecode / UnsupportedImageFeatureException</code> pattern.
USE-02	All public APIs are well documented	Fail	See Findings F2 and F3: the public <code>Fonts.SystemFontCatalog</code> class (a README-advertised headline feature) has no API Reference entry anywhere in the User Guide, and <code>TrueTypeFont</code> 's <code>public IsBold / IsItalic / IsFixedPitch</code> properties and <code>GetNameInfo()</code> / <code>FontNameInfo</code> (also demonstrated in the README's own usage sample) are omitted from the User Guide's <code>TrueTypeFont</code> API Reference section.

1.2.8 2.8 Test Checks

Applicable: No

Reason: The review-set contains no test code files.

1.2.9 2.9 Security Checks

Applicable: No

Reason: The review-set contains no source code files, and neither file makes security claims requiring independent verification beyond the documentation-accuracy checks already covered under section 2.3 (e.g., the description of PDF password/encryption handling is checked for technical accuracy there, not here).

1.2.10 2.10 Code Readability Checks

Applicable: No

Reason: The review-set contains no source code files.

1.2.11 2.11 Requirements vs Documentation Checks

Applicable: No

Reason: The review-set contains no requirements files.

1.2.12 2.12 Requirements vs Implementation Checks

Applicable: No

Reason: The review-set contains no requirements files or source code files.

1.2.13 2.13 Requirements vs Testing Checks

Applicable: No

Reason: The review-set contains no requirements files or test code files.

1.2.14 2.14 Code vs Design Documentation Checks

Applicable: No

Reason: The review-set contains no source code files or design documentation files.

1.3 3. Conclusion

1.3.1 3.1 Summary of Findings

#	Check	Finding
F1	DOC-01	High confidence. docs/user_guide/introduction.md , in the PdfDocument.Render(int, int, int) Exceptions list (around line 1242), states: “ <i>UnsupportedImageFeatureException</i> : <i>Thrown for a well-formed but unsupported color space, stream filter, FormXObject, font subtype (/Type0 / /Type1 / /MMType1 / /Type3), ...</i>” — implying /Type0 , /Type1 , and /Type3 fonts are all unsupported. This directly contradicts: (a) the current implementation (src/DemaConsulting.CanvasNet.Pdf/PdfDocument.Fonts.cs), where BuildResolvedFont explicitly resolves /TrueType , /Type1 , /Type0 (composite, /Identity-H / CIDFontType2), and /Type3 fonts, throwing pdf-font-subtype-{subtype} only for anything else (e.g. /MMType1); and (b) the User Guide’s own preceding prose in the very same PdfDocument section, which lists the actual current scope boundaries as “ /MMType1 fonts, mesh shadings ...” — not Type0/Type1/Type3. This is both a technical inaccuracy and an internal (self-)inconsistency within the file under review, not merely staleness relative to an external document. Recommendation: Correct the Render Exceptions bullet to read /MMType1 only (or otherwise match the method’s actual documented scope boundaries), and recheck the rest of the User Guide/README for any other leftover references to the pre-Phase-9 font-subtype boundary.
F2	USE-02 / DOC-02	High confidence. Fonts.SystemFontCatalog (src/DemaConsulting.CanvasNet/Fonts/SystemFontCatalog.cs) is a public static class with a public SystemFontInfo record struct, a public Fonts property, and public

#	Check	Finding
		<code>FindBestMatch</code> / <code>LoadBundledFallback</code> methods. <code>README.md</code> 's Features list calls it out by name as a headline feature (" <code>System Font Discovery</code> — <code>Fonts.SystemFontCatalog</code> discovers fonts installed on the host operating system, best-effort matches a requested family/style against them, and provides a bundled ... fallback font"), but the User Guide's API Reference section has no "SystemFontCatalog" heading or any documentation of its public surface at all (confirmed by a full heading scan of the file). A reader following the User Guide alone has no way to call this advertised, public feature directly. Recommendation: Add a <code>SystemFontCatalog</code> subsection to the User Guide's API Reference (alongside <code>TrueTypeFont</code>), documenting <code>Fonts</code> , <code>FindBestMatch</code> , <code>LoadBundledFallback</code> , and <code>SystemFontInfo</code> .
F3	USE-02 / DOC-02	High confidence. <code>TrueTypeFont</code> (<code>src/DemaConsulting.CanvasNet/Fonts/TrueTypeFont.cs</code>) has public <code>IsBold</code> , <code>IsItalic</code> , <code>IsFixedPitch</code> properties, a public <code>GetNameInfo()</code> method, and a public <code>FontNameInfo</code> record struct (family/subfamily/full/PostScript name) — all demonstrated directly in <code>README.md</code> 's own usage sample ("Querying a font's name and style metadata"). The User Guide's <code>TrueTypeFont</code> API Reference section (<code>docs/user_guide/introduction.md</code> , ~lines 1257-1326) lists the class's public surface explicitly as a code block, but that code block omits all five of these members entirely; the accompanying prose likewise never mentions name/style querying. Recommendation: Add <code>IsBold</code> / <code>IsItalic</code> / <code>IsFixedPitch</code> , <code>GetNameInfo()</code> , and <code>FontNameInfo</code> to the User Guide's

#	Check	Finding
		<code>TrueTypeFont</code> code-block API listing and prose, mirroring the README's own example.
F4	(Observation — not a Fail)	Medium confidence. README.md 's " PDF Document" feature bullet and its PDF usage sample do not mention password-protected/encrypted-PDF support at all, even though <code>PdfDocument.Open</code> has a public <code>password</code> parameter and the User Guide documents RC4/AES-128/AES-256 decryption (including owner/user password handling) in substantial, accurate detail. This is not a contradiction (README's feature list is intentionally terse and does not claim encryption is <i>unsupported</i>), but it is a completeness gap for a substantial, headline-worthy capability that both files otherwise take care to surface (e.g. shading/tiling patterns, CCITTFax, CalRGB are all at least implicitly covered by the "device color"/"image XObjects" wording, but encryption gets no mention anywhere in README). Recommendation: Consider adding a short clause (e.g. "... and transparently opens common password-protected documents") to the README's PDF feature bullet.
F5	(Observation — not a Fail)	Low confidence. Unlike SVG rendering (Example 13) and every other major feature, PDF rendering has no dedicated numbered "Example" in the User Guide's Examples section (Examples 1-13) — only the brief inline snippet inside the <code>PdfDocument</code> API Reference subsection. This is a minor structural inconsistency in an otherwise example-rich document, not an inaccuracy. Recommendation: Consider adding a dedicated "Example 14: Rendering a PDF Page" for parity with the other features' treatment.

1.3.2 3.2 Overall Outcome

Overall Outcome: Fail

The review-set fails on DOC-01, DOC-02, and USE-02. The most significant issue (F1) is a direct, verifiable technical inaccuracy in the User Guide that also self-contradicts the surrounding text in the same document — exactly the kind of coherence defect this review-set exists to catch. Findings F2 and F3 are completeness gaps: two public, README-advertised API surfaces (`SystemFontCatalog` , and `TrueTypeFont` 's name/style members) have no corresponding User Guide API Reference coverage at all. Findings F4 and F5 are lower-severity observations recorded for the project record but do not themselves change the overall outcome.

Required follow-up before this review-set can be considered Pass/approved:

1. Correct the `PdfDocument.Render` Exceptions entry in the User Guide to accurately reflect that only `MMType1` (among Type1-family/simple-font subtypes) remains unsupported, matching the surrounding prose and the actual implementation (F1).
2. Add User Guide API Reference coverage for `Fonts.SystemFontCatalog` (F2).
3. Add the missing `IsBold` / `IsItalic` / `IsFixedPitch` / `GetNameInfo()` / `FontNameInfo` members to the User Guide's `TrueTypeFont` API Reference (F3).
4. Optionally address F4 and F5 for completeness/parity.

No code or documentation was modified by this review; all findings are reported here for the orchestrator/maintainers to action.